Logical Addressing

IPv4 Addresses & IPv6 Addresses



IPv4 Addresses



IPv4 Addressing

- An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- Two devices on the Internet can never have the same address at the same time.
- The IPv4 addresses are unique and universal.
- IPv4 uses **32-bit addresses**, the address space of IPv4 is **2^{32} or 4,294,967,296.**

Contd…

- Notations
- Classful Addressing
- Classless Addressing
- Network Address Translation (NAT)

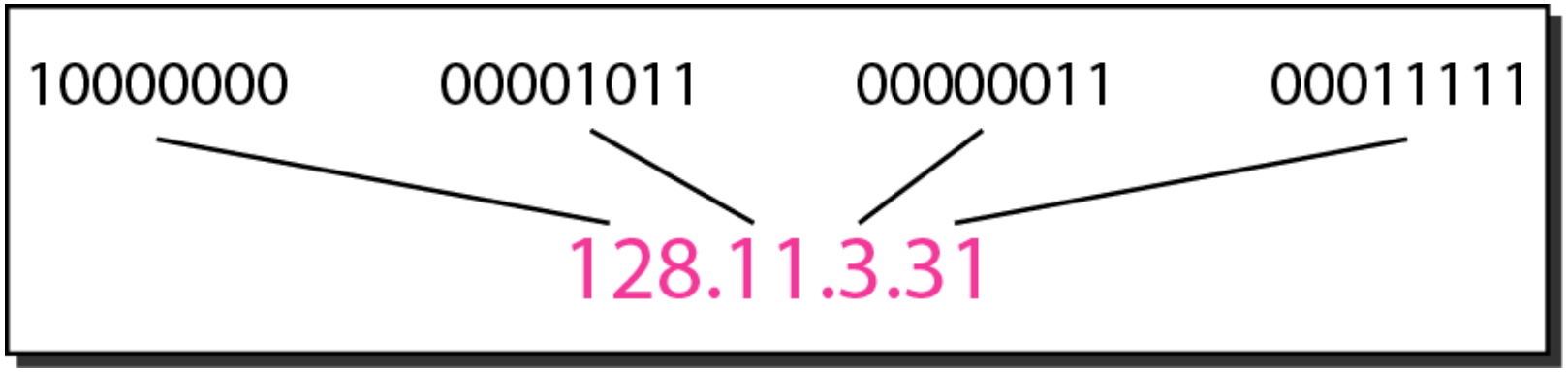
Notations



Notations

- Two prevalent notations to show an IPv4 address are
  - **Binary notation and Dotted decimal notation**
- **Binary Notation:** In binary notation, the IPv4 address is displayed as 32 bits. Each octet is often referred to as a byte.
- So, it is common to hear an IPv4 address referred to as a 32-bit address or a 4-byte address.
- **Dotted-Decimal Notation:** To make the IPv4 address more compact and easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes.

Dotted-decimal notation and binary notation for an IPv4 address



Example

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78
b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent

a.01101111 00111000 00101101 01001110
b. 11011101 00100010 00000111 01010010

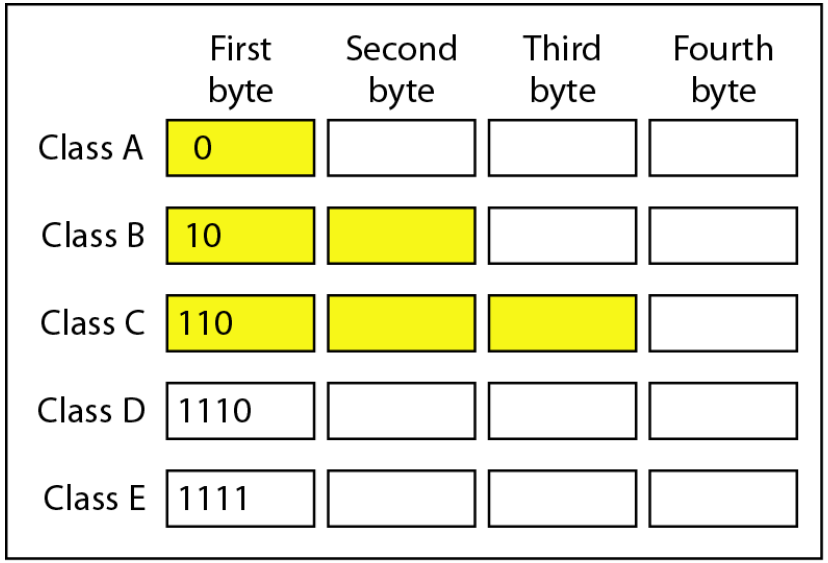
Classful addressing



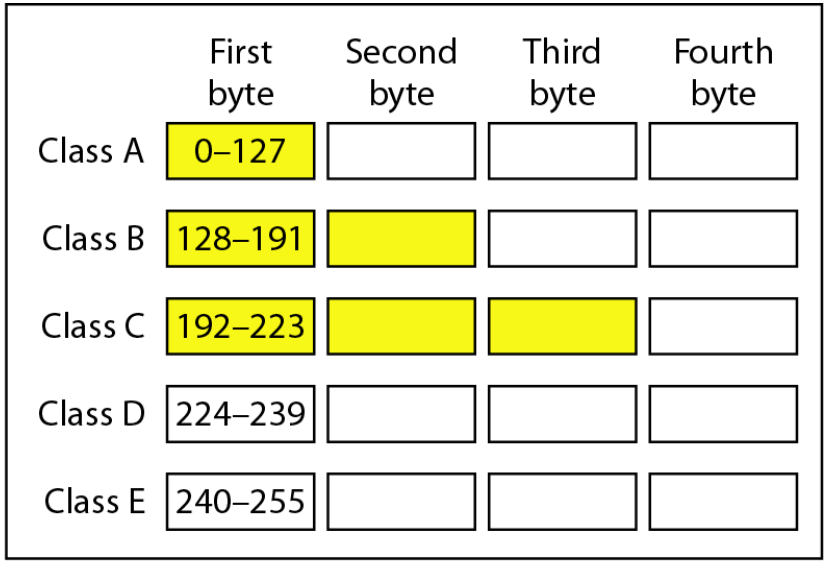
Classful Addressing

- IPv4 addressing, at its inception, used the concept of classes.
- This architecture is called classful addressing. Although this scheme is becoming obsolete.
- In classful addressing, the address space is divided into five classes: A, B, C, D, and E.
- Each class occupies some part of the address space

Finding the classes in binary and dotted-decimal notation



a. Binary notation



b. Dotted-decimal notation

Example

Find the class of each address.

a. 0**0000001** 00001011 00001011 11101111

b. 110**00001** 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111

Solution

a. The first bit is 0. This is a class A address.

b. The first 2 bits are 1; the third bit is 0. This is a class C address.

c. The first byte is 14; the class is A.

d. The first byte is 252; the class is E.

Number of blocks and block size in classful IPv4 addressing

Class	*Number of Blocks*	*Block Size*	*Application*
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

In classfull addressing, a large part of the available addresses were wasted.

Default masks for classful addressing

Class	*Binary*	*Dotted-Decimal*	*CIDR*
A	**11111111** 00000000 00000000 00000000	**255**.0.0.0	/8
B	**11111111 11111111** 00000000 00000000	**255.255**.0.0	/16
C	**11111111 11111111 11111111** 00000000	**255.255.255**.0	/24

Netid and Hostid

- In classful addressing, an IP address in class A, B, or C is divided into netid and hostid.
- These parts are of varying lengths, depending on the class of the address. The netid is in color, the hostid is in white.
- The concept does not apply to classes D and E.
  - In class A, one byte defines the netid and three bytes define the hostid.
  - In class B, two bytes define the netid and two bytes define the hostid.
  - In class C, three bytes define the netid and one byte defines the hostid.

Mask

- Although the length of the netid and hostid (in bits) is predetermined in classful addressing, use a mask (also called the default mask), a 32-bit number made of contiguous 1s followed by contiguous 0s.
- The masks for classes A, B, and C are shown in Table.
- The concept does not apply to classes D and E.

Class	*Binary*	*Dotted-Decimal*	*CIDR*
A	**11111111** 00000000 00000000 00000000	**255**.0.0.0	/8
B	**11111111 11111111** 00000000 00000000	**255.255**.0.0	/16
C	**11111111 11111111 11111111** 00000000	**255.255.255**.0	/24

Contd…

- Classful addressing, which is almost obsolete, is replaced with classless addressing.
- The mask can help us to find the netid and the hostid.
- For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the netid; the next 24 bits define the hostid.
- The last column of Table shows the mask in the form 1n where n can be 8, 16, or 24 in classful addressing.
- This notation is also called **slash notation or Classless Interdomain Routing (CIDR) notation.**

Subnetting

- If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets) or, in rare cases, share part of the addresses with neighbors.
- Subnetting increases the number of 1s in the mask.

Supernetting

- The time came when most of the class A and class B addresses were depleted; however, there was still a huge demand for midsize blocks.
- The size of a class C block with a maximum number of 256 addresses did not satisfy the needs of most organizations.
- Even a midsize organization needed more addresses. One solution was supernetting.
- In supernetting, an organization can combine several class C blocks to create a larger range of addresses.
- In other words, several networks are combined to create a supernetwork or a supemet.

Contd…

- An organization can apply for a set of class C blocks instead of just one.
- For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks.
- The organization can then use these addresses to create one supernetwork.
- Supernetting decreases the number of 1s in the mask.
- For example, if an organization is given four class C addresses, the mask changes from /24 to /22.

Classless Addressing



Address Blocks

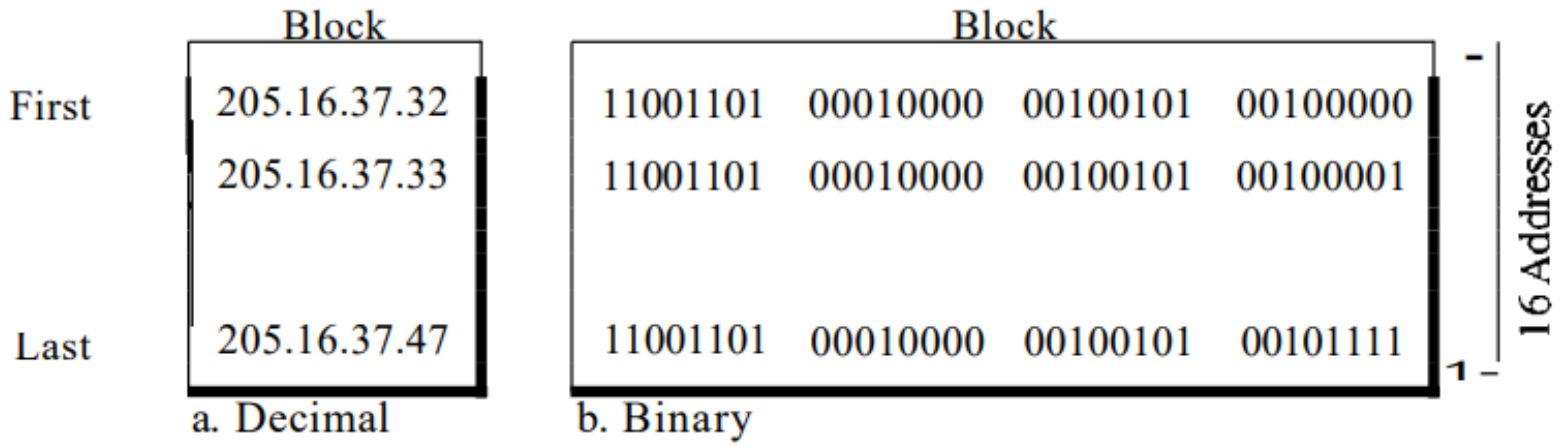
- In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses.
- The size of the block (the number of addresses) varies based on the nature and size of the entity.
- An ISP, as the Internet service provider, may be given thousands or hundreds of thousands based on the number of customers it may serve.

Contd…

Three Restriction:

- The addresses in a block must be contiguous, one after another.
- The number of addresses in a block must be a power of 2 (1, 2, 4, 8, …).
- The first address must be evenly divisible by the number of addresses.

A block of 16 addresses granted to a small organization



Mask

Note

In IPv4 addressing, a block of addresses can be defined as x.y.z.t /*n* in which x.y.z.t defines one of the addresses and the /*n* defines the mask.

- The address and the 1n notation completely define the whole block (the first address, the last address, and the number of addresses).
- **First Address:** The first address in the block can be found by setting the 32 - n rightmost bits in the binary notation of the address to Os.
- **Last Address:** The last address in the block can be found by setting the 32 - n rightmost bits in the binary notation of the address to 1s
- **Number of Addresses:** The number of addresses in the block is the difference between the last and first address. It can easily be found using the formula 2^{32-n}

Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32−28 rightmost bits to 0, we get

11001101 00010000 00100101 0010000

or

205.16.37.32.

Example

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 − 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

Example

Find the number of addresses in Previous Example.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.

$2^{32-n} = 2^{32-28} = 2^4 = 16$ number of addresses

2^{32-n} (n = 28) { — first, — last

Example

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Previous Example /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

a. The first address

b. The last address

c. The number of addresses.

(continued)

Solution

a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.

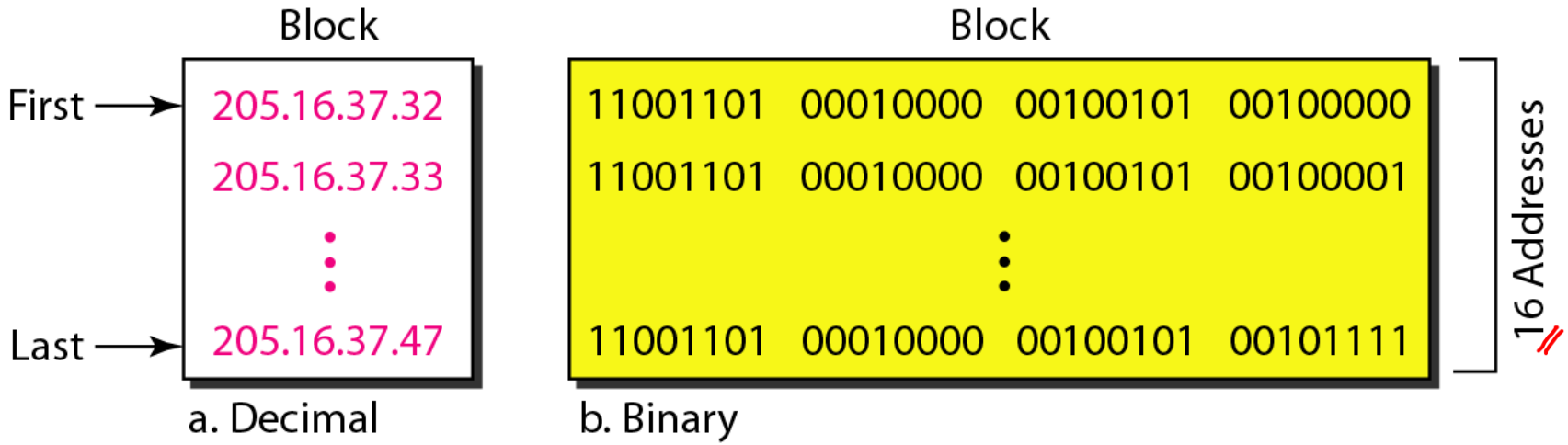
Address:	11001101	00010000	00100101	00100111
Mask:	**11111111**	**11111111**	**11111111**	**11110000**
First address:	11001101	00010000	00100101	00100000

(continued)

b. **The last address can be found by OR ing the given addresses with the complement of the mask. Or ing here is done bit by bit. The result of OR ing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.**

Address:	11001101	00010000	00100101	00100111
Mask complement:	**00000000**	**00000000**	**00000000**	**00001111**
Last address:	11001101	00010000	00100101	00101111

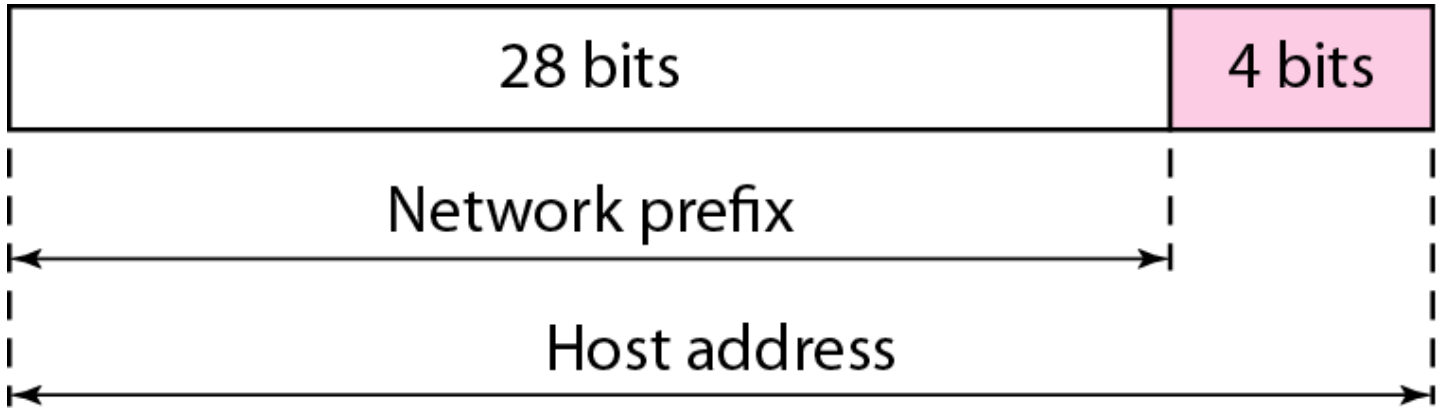
A network configuration for the block 205.16.37.32/28



Note

The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

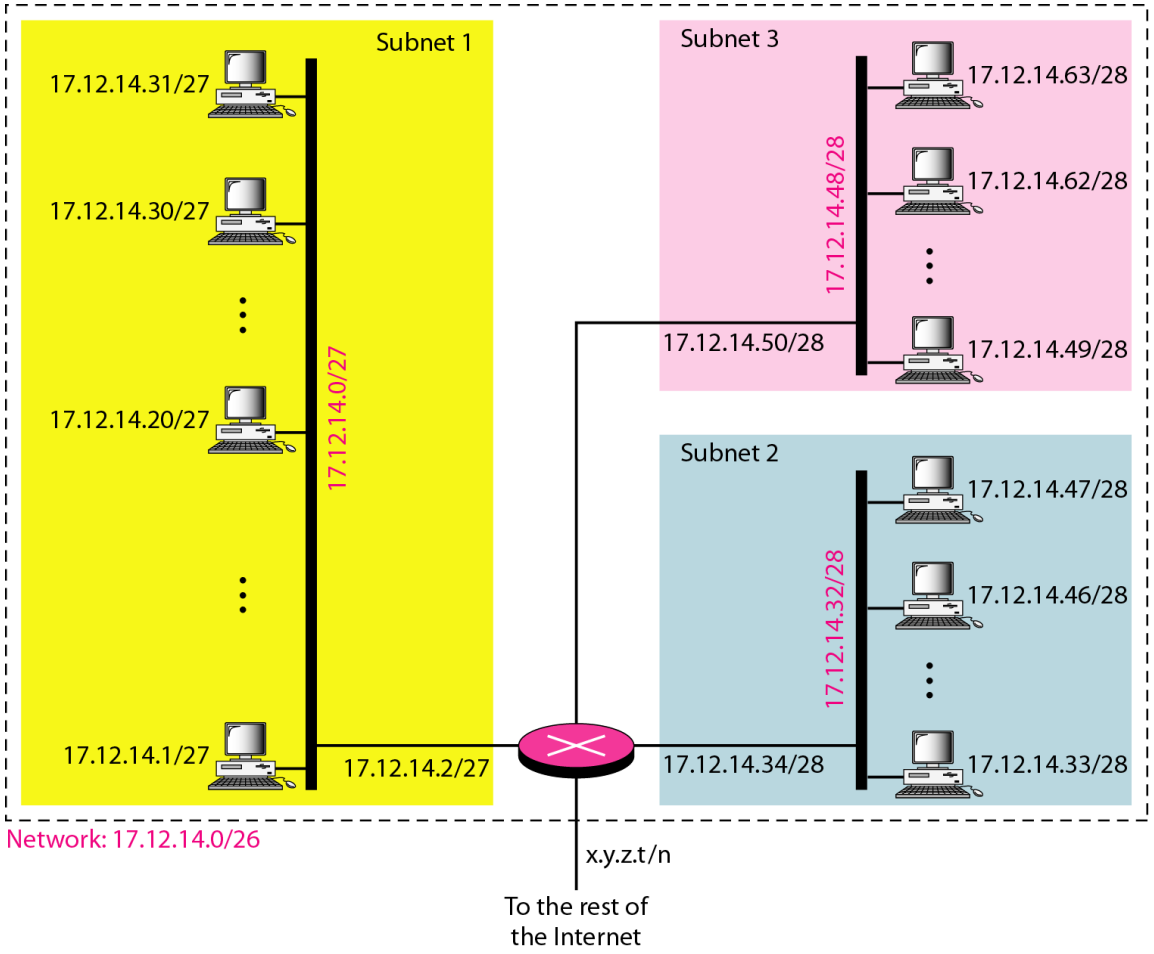
A frame in a character-oriented protocol



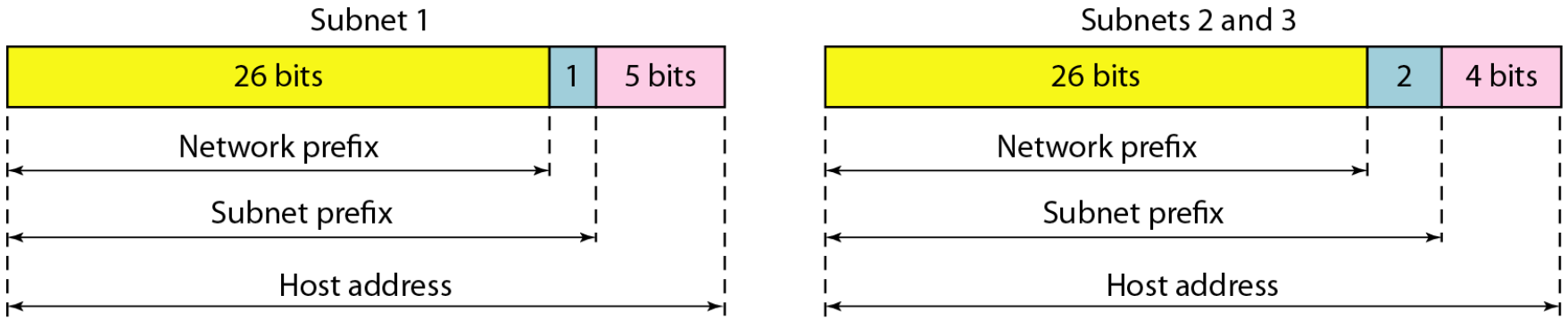
Note

Each address in the block can be considered as a two-level hierarchical structure: the leftmost *n* bits (prefix) define the network; the rightmost 32 − n bits define the host.

Configuration and addresses in a subnetted network



Three-level hierarchy in an IPv4 address



Example

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

a. The first group has 64 customers; each needs 256 addresses.
b. The second group has 128 customers; each needs 128 addresses.
c. The third group has 128 customers; each needs 64 addresses.

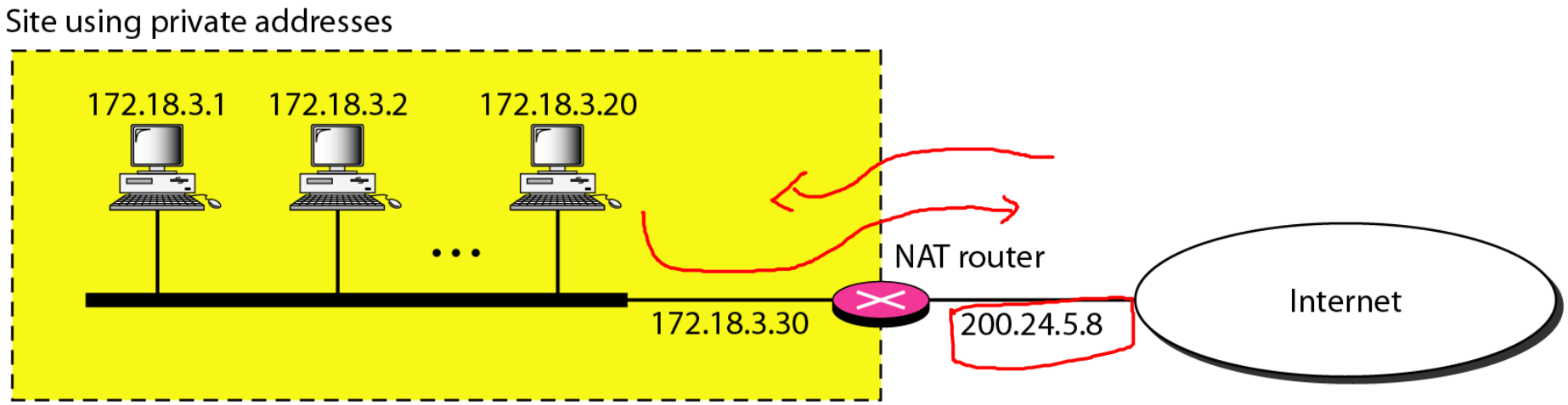
Design the subblocks and find out how many addresses are still available after these allocations.

Network Address Translation (NAT)



- NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally.
- The traffic inside can use the large set; the traffic outside, the small set.
- To separate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved three sets of addresses as **private addresses**.

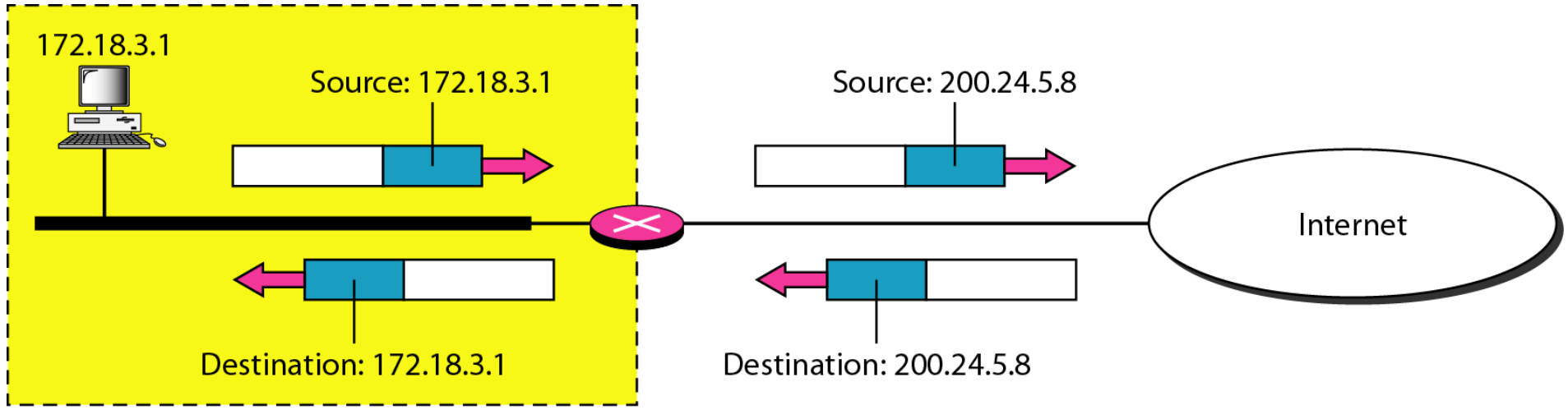
A NAT implementation



Address Translation

- All the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address.
- All incoming packets also pass through the NAT router, which replaces the destination address in the packet (the NAT router global address) with the appropriate private address.

Addresses in a NAT

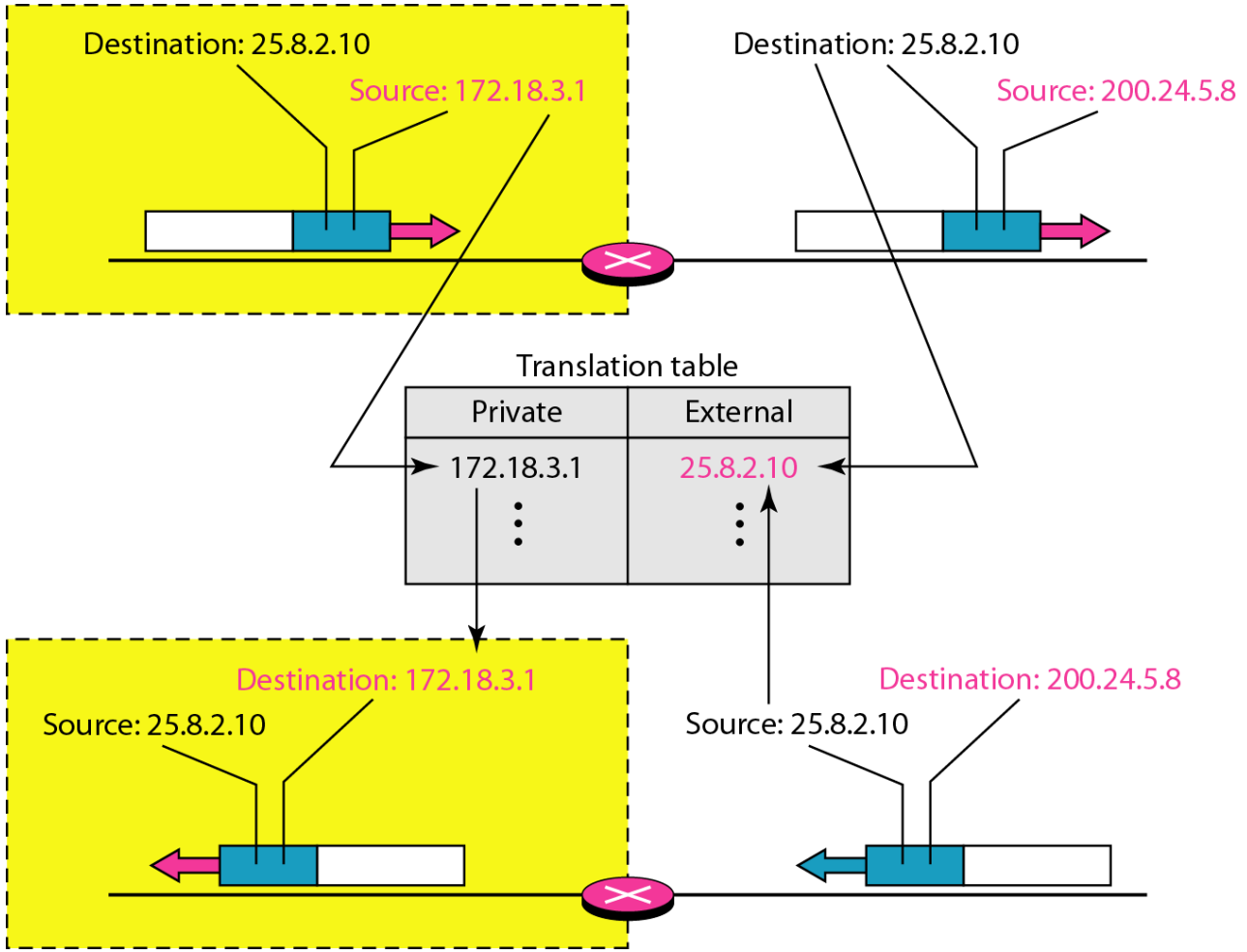


Translation Table

NAT router has a translation table.

- Using One IP Address
- Using a Pool of IP Addresses
- Using Both IP Addresses and Port Numbers

NAT address translation



Five-column translation table

Private Address	*Private Port*	*External Address*	*External Port*	*Transport Protocol*
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
. . .	. . .	. . .	. . .	. . .

IPv6 Addresses



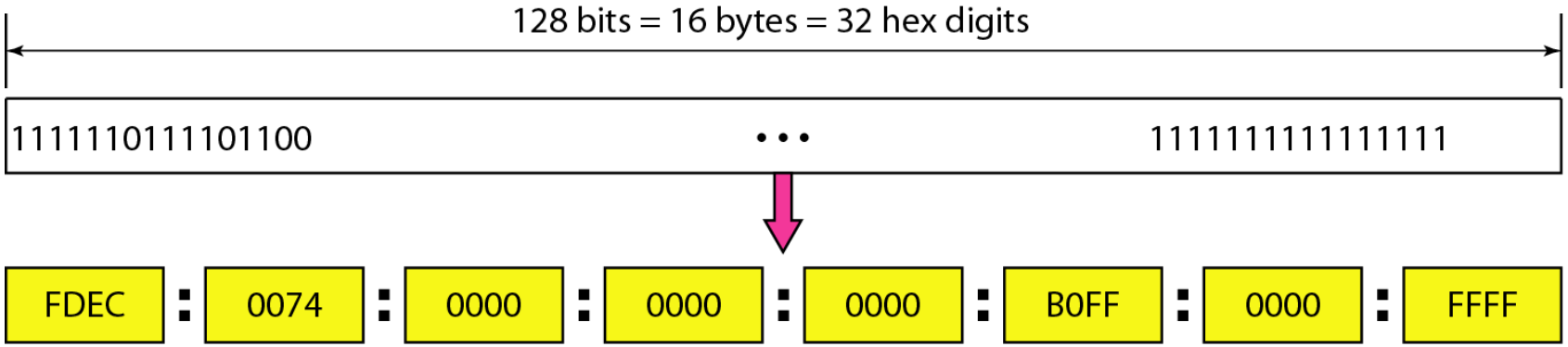
IPv6 ADDRESSES

- Despite all short-term solutions, address depletion is still a long-term problem for the Internet.
- This and other problems in the IP protocol itself have been the motivation for IPv6.
  - **Structure**
  - **Address Space**

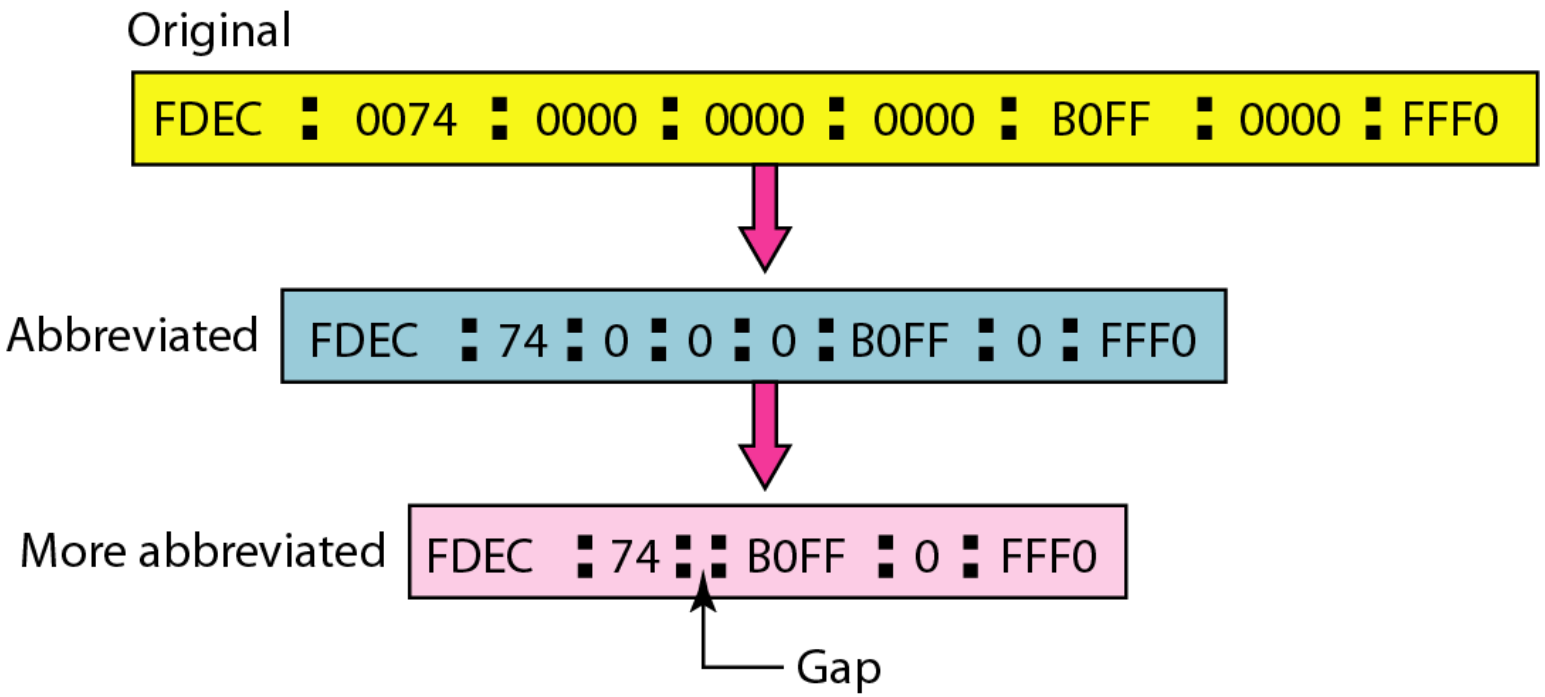
Note

An IPv6 address is 128 bits long.

IPv6 address in binary and hexadecimal colon notation



Abbreviated IPv6 addresses



Example

Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

```
xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx
  0:   15:                  :    1:   12:1213
```

This means that the original address is.

```
0000:0015:0000:0000:0000:0001:0012:1213
```

IPv4 Addressing Vs IPv6 Equivalents

IPv4 Address	**IPv6 Address**
Address Length – 32 bits	128 bits
Address Representation - decimal	hexadecimal
Internet address classes	Not applicable in IPv6
Multicast addresses (224.0.0.0/4)	IPv6 multicast addresses (FF00::/8)
Broadcast addresses	Not applicable in IPv6
Unspecified address is 0.0.0.0	Unspecified address is ::
Loopback address is 127.0.0.1	Loopback address is ::1
Public IP addresses	Global unicast addresses
Private IP addresses (10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16)	Site-local addresses (FEC0::/10)
Autoconfigured addresses (169.254.0.0/16)	Link-local addresses (FE80::/64)

IPv4 vs IPv6 ADDRESSING

Both identify devices on a network, but IPv6 is the next generation with a **much larger address space!**

IPv4 ADDRESSING — 32-bit

1 ADDRESS FORMAT

32-bit address written in dotted decimal notation

`192.168.1.10`

(Four numbers 0 – 255 separated by dots)

2 ADDRESS STRUCTURE

Divided into **Network ID and Host ID**

Network ID	Host ID

Variable Length

3 EXAMPLE 192.168.1.10 /24

192	168	1	10

Network | Host

4 KEY POINTS

- Total addresses: $2^{32} \approx 4.3$ billion
- Short and easy to read
- Addresses are limited

USED IN: Most existing networks, websites, and devices today.

IPv6 ADDRESSING — 128-bit

1 ADDRESS FORMAT

128-bit address written in hexadecimal notation

`2001:0db8:85a3:0000:0000:8a2e:0370:7334`

(Eight groups of 4 hex digits separated by colons)

2 ADDRESS STRUCTURE

Divided into **Network Prefix and Interface ID**

Network Prefix	Interface ID

Typically /64 (First 64 bits) — Last 64 bits

3 EXAMPLE 2001:db8:85a3::8a2e:370:7334 /64

2001	0db8	85a3	0000	0000	8a2e	0370	7334

Network Prefix (First 64 bits) | Interface ID (Last 64 bits)

4 KEY POINTS

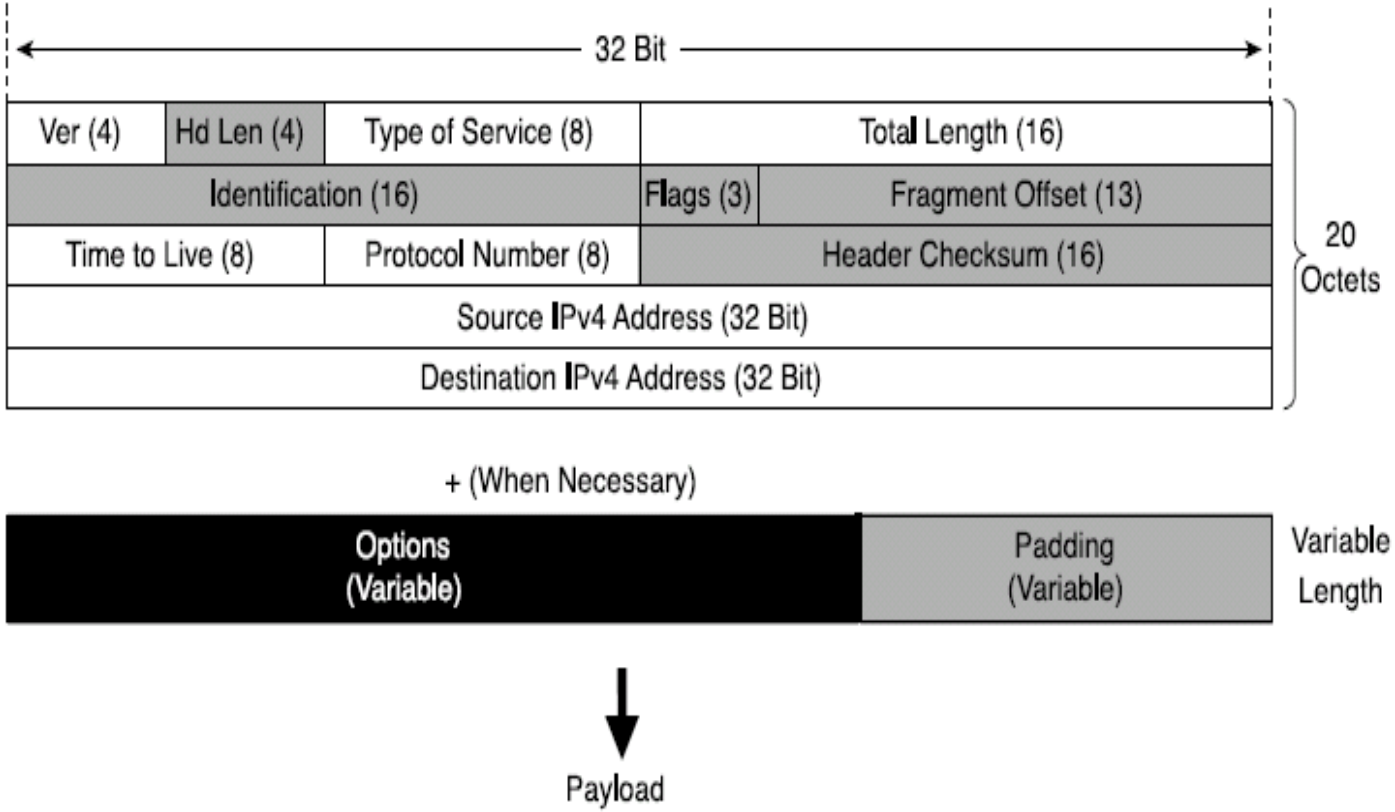
- Total addresses: $2^{128} \approx 3.4 \times 10^{38}$ (practically unlimited)
- More secure with built-in features
- Supports more devices and future growth

USED IN: New networks, mobile networks, IoT, and future applications.

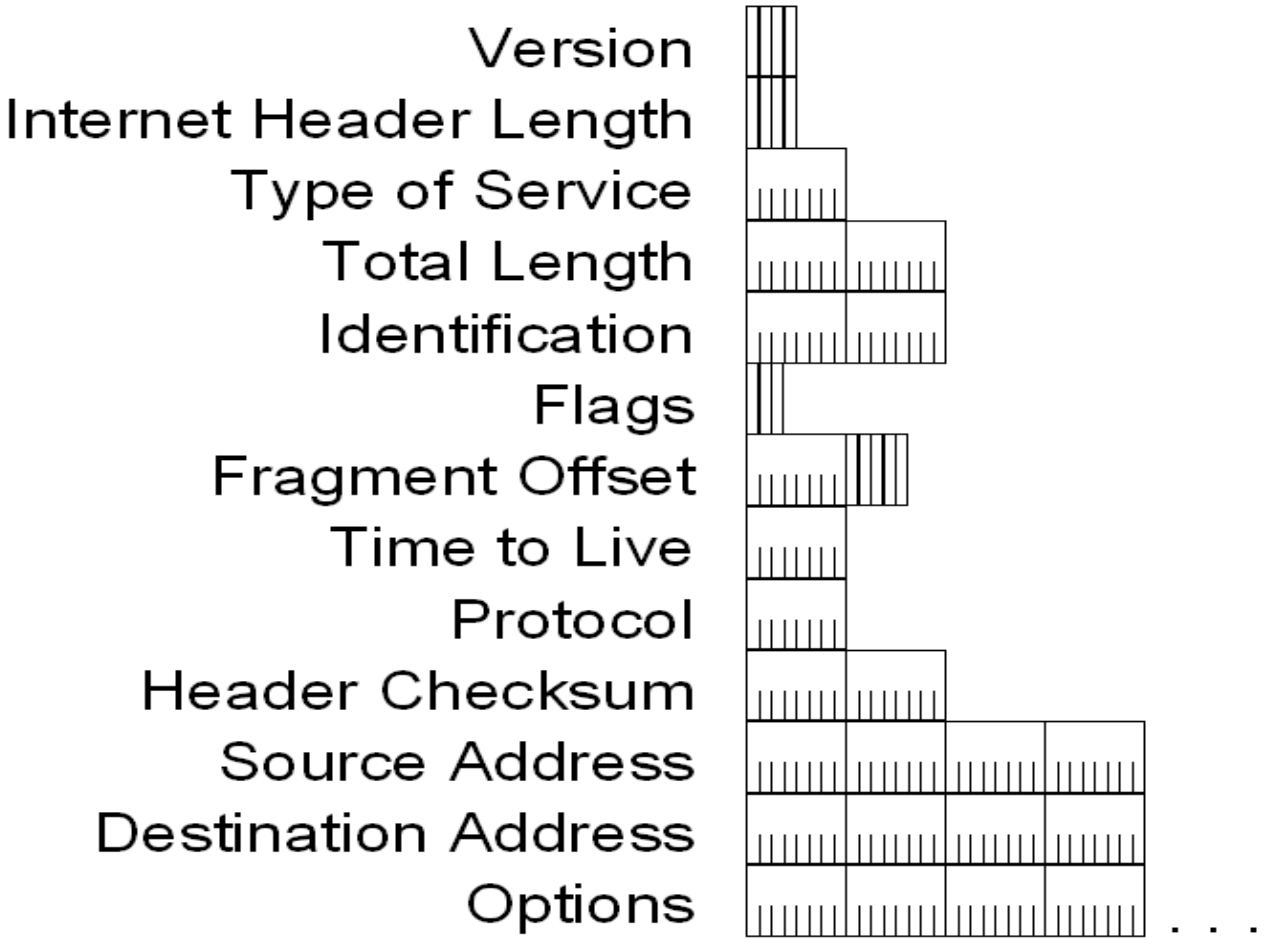
IN SHORT: IPv4 is like a small room with limited seats. | IPv6 is a huge hall with **unlimited** seats for everyone!

IPv4 Header Structure

- Basic IPv4 header contains 12 fields.
- Each field of the IPv4 header has a specific use.
- Shaded field are removed in IPv6.



IPv4 Header - Review



IPv4 Header - Review

- **Version (4 bits)**
  - Indicates the version of IP and is set to 4.
- **Internet Header Length (4 bits)**
  - Indicates the number of 4-byte blocks in the IPv4 header.
  - Because an IPv4 header is a minimum of 20 bytes in size, the smallest value of the Internet Header Length (IHL) field is 5.
- **Type of Service (4 bits)**
  - Indicates the desired service expected by this packet for delivery through routers across the IPv4 internetwork.

IPv4 Header - Review

- **Total Length (16 bits)**
  - Indicates the total length of the IPv4 packet (IPv4 header + IPv4 payload) and does not include link layer framing.
- **Identification (16 bits)**
  - Identifies this specific IPv4 packet.
  - The Identification field is selected by the originating source of the IPv4 packet. If the IPv4 packet is fragmented, all of the fragments retain the Identification field value so that the destination node can group the fragments for reassembly.

IPv4 Header - Review

- **Flags (3 bits)**
  - Identifies flags for the fragmentation process.
  - There are two flags—one to indicate whether the IPv4 packet might be fragmented and another to indicate whether more fragments follow the current fragment.
- **Fragment Offset (13 bits)**
  - Indicates the position of the fragment relative to the original IPv4 payload.

IPv4 Header - Review

- **Time to Live (8 bits)**
  - Indicate the maximum number of links on which an IPv4 packet can travel before being discarded.
  - Originally used as a time count with which an IPv4 router determined the length of time required (in seconds) to forward the IPv4 packet, decrementing the TTL accordingly. When the TTL equals 0,an ICMP Time Expired-TTL Expired in Transit message is sent to the source IPv4 address and the packet is discarded.
- **Protocol (8 bits)**
  - Identifies the upper layer protocol.
  - For example, TCP uses a Protocol of 6, UDP uses a Protocol of 17, and ICMP uses a Protocol of 1.
  - The Protocol field is used to demultiplex an IPv4 packet to the upper layer protocol.

IPv4 Header - Review

- **Header Checksum (16 Bits)**
  - Provides a checksum on the IPv4 header only.
  - The IPv4 payload is not included in the checksum calculation as the IPv4 payload and usually contains its own checksum..
- **Source Address (32 bits)**
  - Stores the IPv4 address of the originating host.
- **Destination Address (32 bits)**
  - Stores the IPv4 address of the destination host.
- **Options (multiple of 32 bits)**
  - Stores one or more IPv4 options.

IPv4 vs IPv6 HEADER

★ Different headers, same goal – deliver data from source to destination!

IPv4 HEADER

20 – 60 bytes (Variable)

Version (4 bits)	IHL (4 bits)	Type of Service (8 bits)	Total Length (16 bits)
Identification (16 bits)		Flags (3 bits)	Fragment Offset (13 bits)
Time to Live (8 bits)	Protocol (8 bits)	Header Checksum (16 bits)	
Source Address (32 bits)			
Destination Address (32 bits)			
Options (0 – 40 bytes)		Padding	

WHAT IT MEANS

- Includes checksum for error checking
- Supports fragmentation
- More fields for control & options
- Header size varies (20 to 60 bytes)
- More processing at routers

IPv6 HEADER

40 bytes (Fixed)

Version (4 bits)	Traffic Class (8 bits)	Flow Label (20 bits)
Payload Length (16 bits)	Next Header (8 bits)	Hop Limit (8 bits)
Source Address (128 bits)		
Destination Address (128 bits)		

WHAT IT MEANS

- No header checksum (rely on upper layers)
- No fragmentation at routers
- Simpler header for faster processing
- Fixed header size (40 bytes)
- Supports huge address space

KEY DIFFERENCE

IPv6 is simpler, faster and built for the future Internet!

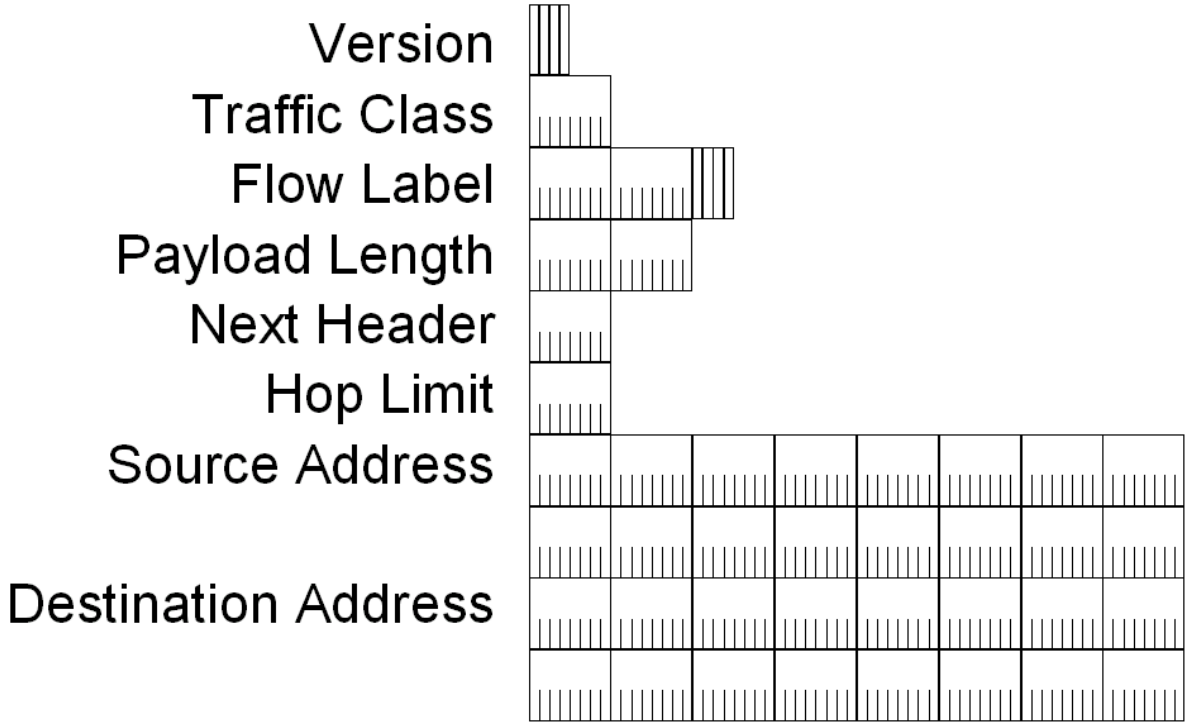
LEGEND (How to read the colors)

- Fields kept from IPv4 to IPv6
- Fields not kept in IPv6
- Name or position changed in IPv6
- New field in IPv6

THE GOAL

Connect every device, everywhere!

IPv6 Header


Version
Traffic Class
Flow Label
Payload Length
Next Header
Hop Limit
Source Address
Destination Address

IPv6 Header Fields

- Based on these rules, RFC 2460 defines the following IPv6 header fields:

1. **Version (4 bits)**
   - 4 bits are used to indicate the version of IP and is set to 6
2. **Traffic Class (8 bits)**
   - same function as the Type of Service field in the IPv4 header.
3. **Flow Label (20 bits)**
   - identifies a flow and it is intended to enable the router to identify packets that should be treated in a similar way without the need for deep lookups within those packets.
   - set by the source and should not be changed by routers along the path to destination.

IPv6 Header Fields

4. **Payload Length (16 bits)**
   - With the header length fixed at 40 bytes, it is enough to indicate the length of the payload to determine the length of the entire packet.
5. **Next Header (8 bits)**
   - Indicates either the first extension header (if present) or the protocol in the upper layer PDU (such as TCP, UDP, or ICMPv6).
6. **Hop Limit (8 bits)**
   - In IPv6, the IPv4 TTL was appropriately renamed Hop Limit because it is a variable that is decremented at each hop, and it does not have a temporal dimension.

IPv6 Header Fields

7. **Source IPv6 Address (128 bits)**
   - Stores the IPv6 address of the originating host.
8. **Destination IPv6 Address (128 bits)**
   - Stores the IPv6 address of the current destination host.

Values of the Next Header Field

Value (in decimal)	Header
0	Hop-by-Hop Options Header
6	TCP
17	UDP
41	Encapsulated IPv6 Header
43	Routing Header
44	Fragment Header
46	Resource ReSerVation Protocol
50	Encapsulating Security Payload
51	Authentication Header
58	ICMPv6
59	No next header
60	Destination Options Header